

Repairing Mission Rewards for Cellular UAV Control: A Factorial Study on a Barcelona Radio Map

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Abstract—A controller can improve communication metrics while failing to complete its flight. We investigate this failure mode in a documented reimplementation of a cellular UAV problem using a received Barcelona ray tracing map with 133 base stations and 17.5 million spatial locations. A four treatment experiment separates replacement of a binary approach reward from termination on safe arrival. All treatments share full observations, continuous acceleration, masked handover actions, and proximal policy optimization. Across five training seeds and 200 held out routes per seed, the legacy reward achieves 21.4% mission success, reward replacement alone achieves 100.0%, and the combined change achieves 99.7%. Termination alone achieves 0.0%. The combined change improves success by 78.3 percentage points, with a paired 95% bootstrap interval of [66.2, 89.5]. On longer routes, reward replacement alone achieves 100.0%, whereas the combined treatment achieves 36.1%. A deterministic reference completes all routes in both sets. The results support repairing the learning objective and show that arrival termination is not by itself a sufficient remedy. They do not establish reproduction of the unavailable original simulator, superiority to classical control, or generalization to other cities.

Index Terms—UAV, cellular connectivity, reward shaping, handover, reinforcement learning, mission completion, reproducibility

I. INTRODUCTION

A 2026 master’s thesis constructed a Barcelona environment using deterministic ray tracing and real base station locations [1, pp. 6–7, Sec. 3.1]. Thesis page locators in this paper refer to printed pages; the PDF viewer page is two greater. Its reward adds a fixed bonus whenever distance decreases [1, p. 12, Eqs. (13)–(14); p. 15, Table 3], and episodes continue after arrival to encourage braking [1, p. 12, Sec. 5.2]. The thesis reports wandering and prioritization of connectivity over task completion [1, p. 19, Sec. 7.2; p. 21, Sec. 8]. Its listed observation vector omits velocity despite acceleration control [1, p. 12, Sec. 5.2; p. 9, Eq. (2)]. These observations motivate a formulation audit, but the written description alone cannot identify the behavior of unavailable implementation code.

Cellular UAV control couples motion with network association. A path changes received powers and cell boundaries, while network decisions change outage, buffering, and handover costs. Consequently, success requires an explicit mission definition as well as favorable communication measurements. Related work studies these objectives jointly, including cargo UAV trajectory planning and cell association [2].

The present study asks a narrower, testable question: within a validated shared environment on the received radio map, does changing the stated reward restore mission completion, and what additional effect does success termination have? We use a factorial design rather than simultaneously changing the learning algorithm and attributing all gains to it. Native hybrid control is an established approach [3]; the same hybrid policy and repaired observation interface are used in every treatment here.

The contribution is an empirical diagnosis with retained artifacts: a source data audit, a controlled four treatment comparison with five training seeds, paired evaluations on standard and longer routes, and an explicit deterministic reference. No new reward shaping theorem or superior general purpose algorithm is claimed. The original source repository and checkpoints remain unavailable; this paper therefore reports a reimplementation study, not a reproduction or an improvement over the thesis’s actual trained policy. Only reward replacement and training termination vary within our factorial comparison. Other differences from the thesis are shared across all arms and are not independently ablated. Tables V and VI give the explicit comparison, including unchanged values and original settings that are not specified.

II. DATA AND SHARED ENVIRONMENT

A. Received Radio Map

The file `Barcelona_dataset_January.h5` contains an $133 \times 5000 \times 3500$ signed 8 bit RSS array, station identities and coordinates, operator membership, and transmit powers. Its metadata gives 2.1 GHz, 100 m altitude, and nominal 1 m grid resolution. The file size is 2,327,593,160 bytes. We use the 87 station operator corresponding to the thesis’s first operator [1, p. 14, Sec. 6.1]. The station counts, frequency, height, and dimensions match the description [1, p. 6, Sec. 3.1.1; p. 7, Table 2], but exact experiment version identity is not established.

The original stored coordinate vectors span 0–5000 m with 5000 samples and 0–3500 m with 3500 samples. Their spacings are approximately 1.00020004 m and 1.00028580 m. Nearest neighbor lookup uses these vectors; all training and evaluation use native resolution. Values are promoted to floating point before radio arithmetic. The untouched source checksum and schema are retained.

A full scan finds that 27.77245% of all link entries equal -128 , while the other entries range from -110 to -25 dBm. We provisionally interpret -128 as an absent usable path and assign it zero received power. The generator's sentinel and quantization conventions still require confirmation. For the selected operator, every grid position has at least 38 stations above -96 dBm, and the strongest RSS ranges from -51 to -25 dBm. Thus the stored map has no inevitable geographic coverage hole under this RSS threshold. This observation does not guarantee a permissible handover or adequate SINR.

B. Motion, Mission, and Observations

The thesis writes scalar speed and a selected direction in its position update [1, p. 9, Sec. 3.2, Eq. (2)]. Our shared benchmark instead preserves an inertial velocity vector at constant altitude. With velocity projection,

$$\mathbf{v}_{t+1} = \Pi_{\|\mathbf{v}\| \leq v_{\max}}(\mathbf{v}_t + \mathbf{a}_t \Delta t), \quad (1)$$

$$\mathbf{p}_{t+1} = \mathbf{p}_t + \frac{1}{2}(\mathbf{v}_t + \mathbf{v}_{t+1})\Delta t. \quad (2)$$

The policy supplies radial and lateral acceleration components in a frame aligned with the current destination bearing. Their norm is limited to 5 m/s^2 , and speed is limited to 25 m/s . This continuous action interface replaces the five acceleration bins and eight headings [1, p. 12, Sec. 5.2; p. 15, Table 3]. The thesis uses a 100 m/s penalty threshold [1, p. 15, Table 3]; ours is a hard 25 m/s cap. The step is 1 s and the deadline is 200 s . The source gives 1 ms in prose [1, p. 14, Sec. 6.1] and 0.1 s in Table 3 [1, p. 15]; we explicitly choose a new time discretization.

Safe arrival is the common event

$$\mathcal{G}_t = \{\|\mathbf{p}_t - \mathbf{g}\| \leq 10 \text{ m}, \|\mathbf{v}_t\| \leq 2 \text{ m/s}\}. \quad (3)$$

The 10 m position tolerance is retained from Table 3 [1, p. 15]; the numerical 2 m/s terminal speed test is our addition to the stated stopping intention [1, p. 9, Sec. 3.2]. A boundary crossing, battery exhaustion, or five consecutive RSS outage samples ends a failure. The thesis explicitly terminates on energy exhaustion and penalizes boundary, RSS, and speed violations [1, p. 12, Sec. 5.2, Eq. (14)]; our boundary and repeated outage terminations are additional rules. Speed is constrained directly. Building geometry supports propagation in the thesis [1, p. 6, Sec. 3.1.1]. Our received RSS array does not provide a collision model, and obstacle safety is outside the evaluated model; we do not assume the original code enforced it.

Every treatment receives the same 42 dimensional observation: position; destination bearing and distance; radial, lateral, and total velocity; remaining time encoded by elapsed time; energy; buffer; prior arrival flag; resource and load phases; network decision phase; serving SINR; outage count; and candidate RSS, relative positions, identities, and masks. This replaces the listed six component vector of position, serving station, resource group, energy, and buffer [1, p. 12, Sec. 5.2]. Observation repair is shared, and its independent effect is not estimated.

C. Association and Communication Proxies

A categorical policy selects stay or one of the four strongest candidate stations. A request is admissible when candidate RSS exceeds both -96 dBm and serving RSS plus 3 dB . Network actions occur every five seconds, with an additional opportunity on an observed serving RSS outage. This emergency rule was introduced during validation, before training, when a periodic decision could otherwise arrive after the outage failure deadline. We do not implement or claim a complete 3GPP time to trigger procedure. Masking follows the standard practice of excluding invalid categorical actions [4]. The inequality follows the thesis's A3 equation [1, p. 9, Eq. (3)], with the numerical offset from Table 3 [1, p. 15]; the five action candidates and decision interval are our choices. The source greedy controller uses viability counters [1, pp. 13–14, Sec. 5.4], which we do not copy.

There are twelve resource groups and six occupied background groups per station in a fixed spatial pattern. A shared first available allocation rule chooses the resource group; resource allocation is not learned. Scenario phase rotates group labels globally and does not supply independent traffic dynamics. The source permits learned RBG requests [1, p. 9, Sec. 3.3.1; p. 12, Sec. 5.2] and describes random occupancy of at least half the groups [1, p. 14, Sec. 6.1]; both differ from our shared allocator. Since the source RSS is a downlink budget [1, p. 7, Eq. (1)], station powers are aggregated as a downlink interference proxy:

$$P_k = 10^{(\text{RSS}_k - 30)/10}, \quad \text{SINR} = \frac{P_s}{N + \sum_{k \neq s} o_k P_k}. \quad (4)$$

Here powers are in watts, o_k indicates background occupancy of the selected group, and noise including the noise figure is -103.41 dBm . The ideal service rate is $1.44 \cdot 10^6 \log_2(1 + \text{SINR}) \text{ bit/s}$. It is neither measured throughput nor a validated uplink interference model. This replaces the printed bandwidth-plus-logarithm rate, SNR in dB, and sum of RSS values labeled uplink interference [1, p. 10, Eqs. (7)–(9)]. It changes the interference model as well as correcting power arithmetic; we cannot establish whether the original code used the printed expressions.

Deterministic offered traffic is 200 kbit/s , buffer capacity is 1.28 Mbit , and a handover adds 4800 bits . The control overhead and nominal buffer capacity follow Table 3 [1, p. 15], interpreting KB as decimal kilobytes. Deterministic traffic replaces the Poisson formulation [1, pp. 9–10, Eqs. (5)–(6)]; the thesis also gives different data packet sizes in prose and table [1, pp. 14–15, Sec. 6.1, Table 3]. The backlog/service delay retains the ratio in Eq. (4) [1, p. 9], but we explicitly use postservice backlog and censor at 200 s . Energy is the integral of $100 + 0.4\|\mathbf{v}\|^2$ watts against a 100 kJ budget. This explicit proxy replaces the mass, efficiency, lift-to-drag and electronics expression [1, p. 10, Eq. (10)] and the capacity labeled 1000 kW [1, p. 15, Table 3]. It does not claim a calibrated rotorcraft model. Table I summarizes the main departures from the source description.

TABLE I
SHARED REIMPLEMENTATION CHOICES, HELD FIXED ACROSS TREATMENTS

Aspect	Declared benchmark choice
Observations	Goal and velocity observed; 42 dimensions
Motion	Continuous acceleration, 1 s steps, 25 m/s cap
Network	Stay plus four candidates; masks; outage requests
Resources	Common deterministic first available allocator
Interference	Linear downlink cochannel power proxy
Traffic and energy	Deterministic traffic; explicit energy proxy
Source equivalence	Original code and checkpoints unavailable

III. REWARD TREATMENTS AND LEARNING

A. Why the Binary Approach Bonus Is Suspect

Let d_t be destination distance, D_t the declared delay proxy, I_t the linear interference proxy, and h_t a handover indicator. The legacy reward treatment implements the structure in Eqs. (13)–(14) [1, p. 12], with the equal-policy weights, scales, and bonus in Table 3 [1, p. 15]:

$$r_t^L = \frac{.35}{1 + 10D_t} + \frac{.30}{1 + 10^5 I_t} + \frac{.35}{1 + 100h_t} + 12\mathbb{K}[d_{t+1} < d_t] - 10O_t - 10B_t, \quad (5)$$

where O_t and B_t mark RSS outage and a boundary crossing. Energy exhaustion overrides the reward with -10 . The shared speed constraint replaces a soft speed penalty. The interference input is corrected to watts for all treatments; thus this is an ablation of the written incentive, not literal replication of potentially erroneous source equations.

The approach term depends on the sign of progress but not its magnitude. A controller can receive the same bonus for very slow progress as for efficient flight. A retreat carries no matching distance cost, permitting positive approach bonuses on a closed excursion outside the destination region. Moreover, stopping the episode without a compensating mission objective can remove future positive reward and reinforce avoidance of completion. Similar failures motivate potential based shaping [5].

B. Mission Reward and Terminal Potential

Define the potential, with $L = 100$ m,

$$\Phi(s) = -\frac{d + \|\mathbf{v}\|^2 / (2a_{\max})}{L}. \quad (6)$$

The speed term is a stopping distance scale. The replacement reward is

$$r_t^M = 20A_t - 20F_t - 0.05 - 0.05c_t + \gamma\Phi(s_{t+1}) - \Phi(s_t), \quad (7)$$

where A_t marks first safe arrival, F_t a terminal failure before any safe arrival, and $c_t = \min(D_t, 1) + O_t + h_t$. The prior arrival flag is observed, and the success bonus is paid only once even if training continues afterward. Set the potential to

TABLE II
FACTORIAL TRAINING TREATMENTS

Label	Reward	Terminate on arrival
Legacy	r^L	No
Termination only	r^L	Yes
Reward only	r^M	No
Both	r^M	Yes

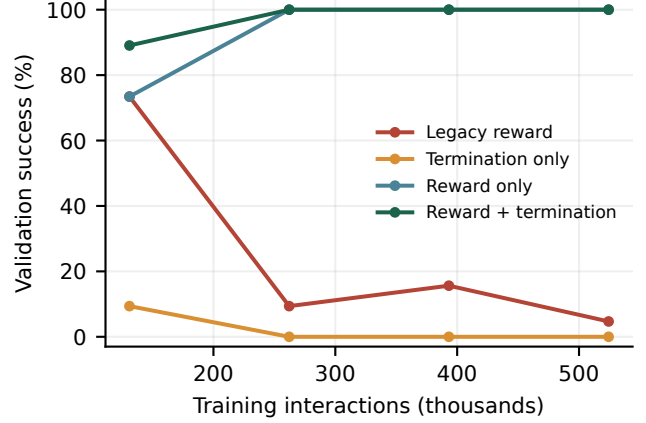


Fig. 1. Pilot seed 42 on 64 validation routes. Curves are used to document development and budget choice, not to substitute for the frozen test comparison.

zero at every terminal, including the deadline. For any episode ending at T , the shaping contribution satisfies

$$\sum_{t=0}^{T-1} \gamma^t [\gamma\Phi(s_{t+1}) - \Phi(s_t)] = -\Phi(s_0). \quad (8)$$

Consequently, within each fixed terminal formulation, shaping preserves the discounted base objective [5]. This identity is checked numerically. It does not imply that the finite weighted base reward certifies every physical constraint or guarantees successful learning. The identity concerns raw rewards. Running normalization and clipping in PPO can change the effective update objective; this experiment tests the declared learning recipe without isolating those interactions.

C. Common PPO Implementation

All arms use proximal policy optimization [6] with separate actor and critic networks, each containing two 64 unit tanh layers. The actor has a Gaussian motion head and a masked categorical association head. Motion samples are transformed through tanh and the acceleration norm limit; likelihood ratios are consistently computed in latent action space. The entropy regularizer sums latent Gaussian and masked categorical entropy.

Training uses 64 parallel lanes, 128 step rollouts, four epochs per rollout, 512 sample minibatches, Adam learning rate $3 \cdot 10^{-4}$, clip 0.2, $\gamma = 0.99$, GAE parameter 0.95, entropy coefficient 0.005, value coefficient 0.5, and gradient norm limit 0.5. Rewards are divided by a running standard deviation of

discounted returns without subtracting a mean, and clipped to $[-10, 10]$, consistently across arms. Finite horizon terminals are not bootstrapped. No pretrained model, demonstrations, or imitation loss is used. The thesis uses Stable Baselines [1, p. 13, Sec. 5.3]; our PPO is a new native implementation. Table VI contrasts every reported source setting with ours [1, p. 16, Table 4]. For example, learning rate increases from $3 \cdot 10^{-5}$ to $3 \cdot 10^{-4}$, entropy weight decreases from 0.01 to 0.005, and the source’s 0.03 target KL is replaced by diagnostic KL logging without an early stopping threshold.

IV. EVALUATION PROTOCOL

A. Validation, Freeze, and Held Out Routes

Training uses 4096 routes sampled with lengths 200–1000 m, with 64 separate validation routes. The final standard set contains 200 new routes of the same length range. A second set contains 200 routes of length 1000–1800 m. These are unseen coordinate pairs in the same city, not independent cities or held out geographic regions. Starts and destinations stay inside documented map margins. All coordinates, seeds, and background phases are saved. The original tuning and comparison sections do not specify three independent training, validation, and test route sets [1, pp. 15–16, Secs. 6.2–6.3]. This is a limit of the written protocol, not proof that its unavailable code lacked such a separation.

Six pilots, using seeds 41 and 42, established learning behavior and the 524,288 interaction budget. All final arms use seeds 1101–1105 and exactly that budget, totaling 10,485,760 interactions. Source hashes and configuration were frozen before trained policy test evaluation. The initial feasibility inspection used separate exploratory route sets, which were retired before training; final route seeds are 42012 and 42013. Final checkpoints are evaluated without early stopping or test based selection. Each lane follows the same ordered training scenario stream across treatments, although successful termination changes how many episodes fit in an equal interaction budget.

The evaluated continuous action is the Gaussian mean after transformation, and the network action is the masked categorical mode. Every policy’s evaluation stops at first safe arrival, independently of its training termination rule. Thus the continuing training treatment is still evaluated as a finite mission.

B. Metrics, Reference, and Uncertainty

Mission success is primary. Recorded feasible success additionally requires no packet overflow and an RSS outage fraction at most 5%. This limited feasibility label does not imply obstacle or airframe certification. Failure counts and final distances are retained. Time, path, energy proxy, delay, interference, SINR, and handovers are reported conditional on success with their sample counts, avoiding aggregation of failed flights into radio gains.

The deterministic reference points its desired velocity toward the goal, uses a braking schedule based on stopping distance, and requests the strongest admissible station. It

shares every environmental constraint and resource rule. It establishes that the tested missions are attainable and provides a practical nonlearning comparison; it is not asserted to be globally optimal.

We show all five seed outcomes and standard deviations. The primary contrast is both changes minus legacy success on standard routes. A 10,000 draw crossed bootstrap resamples the five seed identities and 200 route identities, using the same sampled identities for both arms. This preserves pairing and avoids counting 1000 episodes as 1000 independent training runs. Other factorial and longer route contrasts are secondary. Interval reporting follows the broader need to expose uncertainty in few run RL evaluations [7]. Five seeds still provide limited resolution and uncertain tail behavior.

The prespecified practical gate for a positive repair finding requires at least 90% mean standard mission success, at least 90% recorded feasible success, and a positive lower 95% bootstrap bound for the primary contrast. It is a reporting gate rather than a claim of statistical or deployment certification.

V. RESULTS

A. Completion and Factorial Effects

Table III and Fig. 2 present the full frozen comparison. Reward replacement alone achieves 100.0% standard success compared with 21.4% for the legacy reward. The combined change achieves 99.7%, while termination alone achieves 0.0%. The primary gain is 78.3 percentage points, with a paired bootstrap interval [66.2, 89.5]. Recorded feasible success for the combined change is 97.8%, satisfying the practical gate. These results identify the replacement reward as the essential tested intervention; changing arrival termination alone does not resolve the failure.

The pilots also illustrate why longer training cannot automatically repair a misaligned objective. In pilot seed 42, legacy validation success initially rises to 73.4% and then falls to 4.7% by the final budget. Reward replacement reaches 100% by approximately 262,144 interactions and retains it through the pilot’s final checkpoint. Figure 1 is explicitly exploratory validation evidence from one pilot, not a confidence interval over final seeds.

B. Longer Routes and the Role of Termination

Reward replacement with continuing training reaches 100.0% success on longer routes. The combined treatment reaches 36.1% with seed standard deviation 12.6 percentage points. Thus the treatment that provides efficient completion within the training range does not necessarily generalize best to longer flights. Training termination changes episode exposure and visitation distribution even under equal interaction budgets, but the experiment does not isolate the mechanism causing this extrapolation difference. The longer route weakness is retained rather than removed by further test driven tuning.

Figure 3 shows the first predefined standard test route for seed 1101, selected by index rather than performance. Its trajectories and distance traces illustrate the different learned

TABLE III

FROZEN RESULTS: PERCENTAGES ARE MEAN $\pm$ STANDARD DEVIATION OVER FIVE TRAINING SEEDS. STANDARD ROUTES ARE 200–1000 M; LONGER ROUTES ARE 1000–1800 M. EACH LEARNED POLICY EVALUATES 200 ROUTES PER SPLIT.

Treatment	Mission success (%)		Recorded feasible success (%)		Successful episodes	
	Standard	Longer	Standard	Longer	Standard	Longer
Legacy	21.4 $\pm$ 14.3	0.0 $\pm$ 0.0	20.8 $\pm$ 13.9	0.0 $\pm$ 0.0	214/1000	0/1000
Termination only	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0/1000	0/1000
Reward only	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	98.7 $\pm$ 1.4	96.0 $\pm$ 1.1	1000/1000	1000/1000
Both	99.7 $\pm$ 0.7	36.1 $\pm$ 12.6	97.8 $\pm$ 0.8	35.2 $\pm$ 12.6	997/1000	361/1000
Deterministic reference	100.0	100.0	97.0	95.5	200/200	200/200

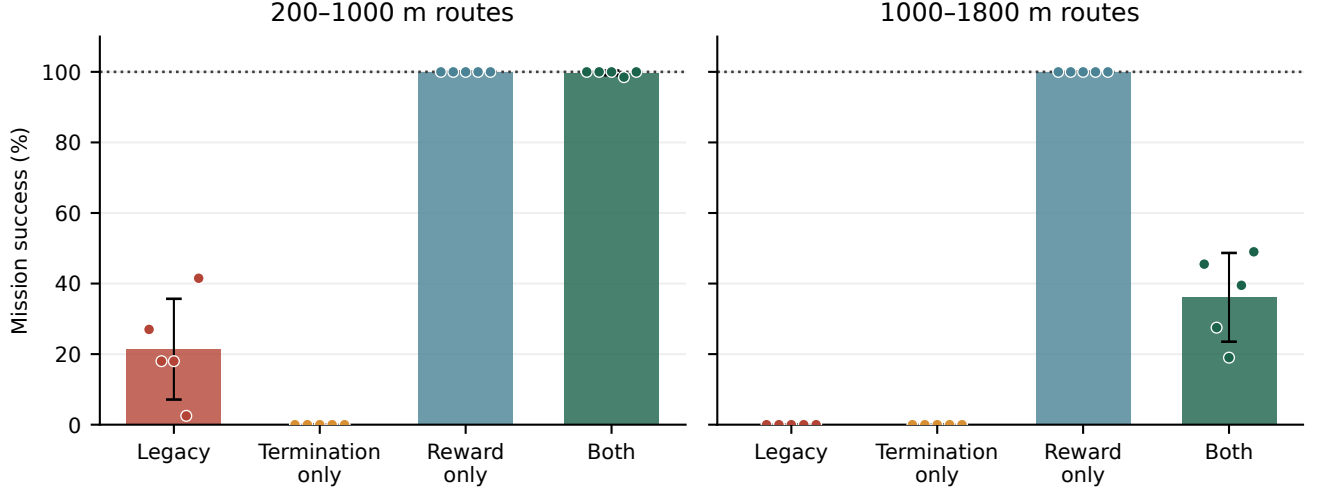


Fig. 2. Frozen test mission success. Each dot is one training seed, bars show the mean, and error bars show seed standard deviation. There are 200 shared routes per split and five seeds per learned treatment. The dotted line is the deterministic reference. No test outcome was used to tune the frozen policies.

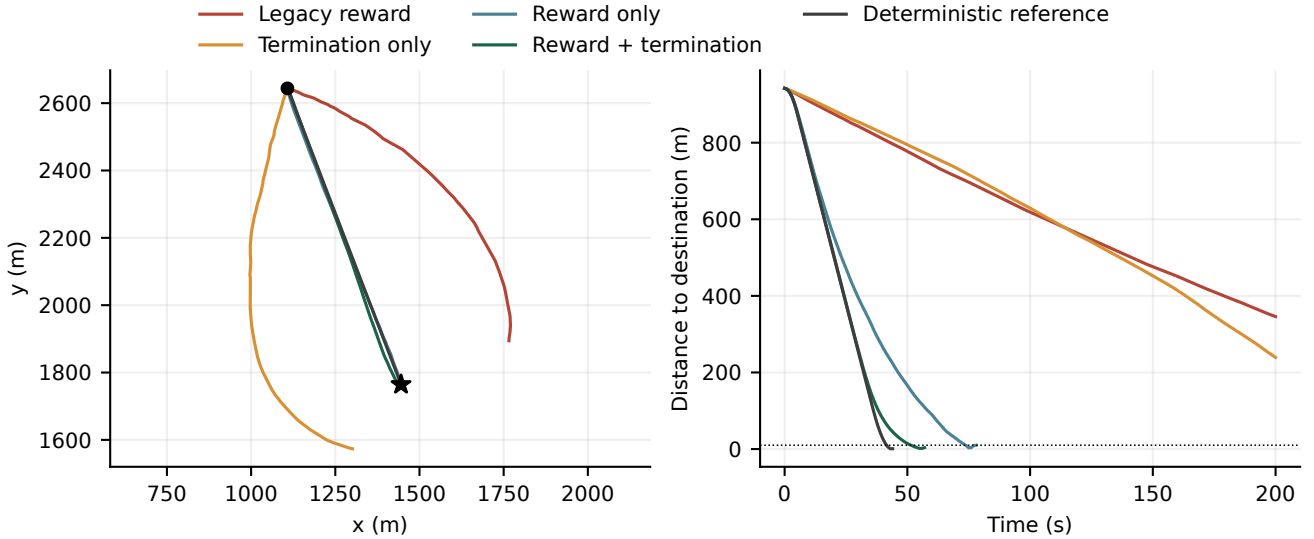


Fig. 3. First standard test route and first declared training seed. The black circle and star denote start and destination. All learned treatments and the deterministic reference share the scenario. The dotted distance line marks 10 m; low speed is also required for success.

behavior, while the aggregate tables remain the basis for the conclusions. Proximity by itself is insufficient: the evaluated success criterion also requires low speed.

C. Efficiency and the Deterministic Reference

The deterministic reference achieves 100.0% standard and 100.0% longer route success. Among successful standard

TABLE IV

STANDARD ROUTE METRICS CONDITIONAL ON MISSION SUCCESS, POOLED ACROSS THE FIVE SEEDS. THE DETERMINISTIC REFERENCE HAS ONE RUN ON 200 ROUTES. DIFFERENT SUCCESSFUL SUBSETS PREVENT UNQUALIFIED SECONDARY DOMINANCE CLAIMS. ENERGY, DELAY, AND INTERFERENCE ARE THE DECLARED PROXIES.

Treatment	n	Time (s)	Energy (kJ)	Handovers	SINR (dB)	Delay (s)	Outage (s)	Interf. (μ W)
Legacy	214	141.25	14.80	8.83	-6.34	0.88	0.07	0.70
Termination only	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Reward only	1000	50.23	8.29	3.96	-7.91	2.58	0.09	0.74
Both	997	36.52	8.17	2.99	-8.13	2.83	0.09	0.74
Deterministic reference	200	28.62	7.99	2.38	-8.51	3.45	0.09	0.74

missions, mean flight times are 141.2 s for legacy, 50.2 s for reward only, 36.5 s for both changes, and 28.6 s for the reference. Because the successful subsets differ, these means alone are not paired efficiency comparisons. On matched successful standard routes, the combined policy takes 7.9 s more time than the reference on average. The evidence establishes repair of a learned objective, not an advantage over the deterministic controller.

Table IV reports success conditional secondary metrics with sample counts. The radio summaries are downlink proxies and the energy measure is uncalibrated. A controller with lower interference or delay on a subset of completed short routes is not ranked above a controller that completes the whole route set. Raw per episode records permit matched subset analysis without concealing failures or mixing physical units. The original results emphasize SNR, outage, interference, remaining energy, and handover count [1, pp. 16–20, Sec. 7, Figs. 5–8]. Our success rate, terminal speed and distance checks, success conditional reporting, and paired seed uncertainty are additional evaluation choices.

VI. VALIDITY AND REPRODUCIBILITY

The initial test suite contains 34 passing checks covering movement, speed and acceleration bounds, position and speed arrival conditions, one time success payment, terminal potential telescoping, finite horizon GAE, mask validity, likelihood consistency, radio axes and spacing, emergency handovers, identical dynamics across reward arms, deterministic completion, and seed repeatability. Reloading the first declared reward only checkpoint reproduces all 200 recorded standard test episode records exactly. Passing these checks validates important properties of this code, not equivalence to unavailable source code.

Experiments use Python 3.12, CPU PyTorch 2.4.1, NumPy 2.3.5, and h5py 3.16.0 on a Windows machine with 16 logical processors and approximately 32 GiB RAM. The sum of recorded training and final validation times is approximately 9.4 minutes, excluding map loading and final test evaluation. Checkpoints, all configurations, route splits, training logs, per episode metrics, source hashes, analysis scripts, and figures are retained locally. The received map is unchanged and excluded from version control. Independent reproduction currently requires obtaining that private artifact from its owner; no public dataset release is claimed.

Several limitations constrain interpretation. First, the shared full state, continuous actions, and network masks mean the results are not a direct comparison with the thesis’s partially observed discrete controller. Second, the radio, scheduling, delay, and energy models include declared simplifications; the full original simulator and exact dataset version remain unverified. Third, training and testing share one city, one height, one frequency, and one fixed load pattern. Fourth, five training seeds cannot establish robust tail behavior. Fifth, the controller is evaluated with deterministic actions and a fixed training budget, so the measured comparison includes these choices. Finally, the deterministic reference is already highly effective; the study does not justify adding a learning controller for this setting on efficiency grounds alone.

VII. CONCLUSION

On the received Barcelona radio map and within an explicit shared reimplement, replacing the binary approach reward restores reliable mission completion in the tested training range. The factorial comparison shows that success termination alone is insufficient, and that its combination with reward repair can reduce longer route generalization relative to reward repair alone. Full mission outcomes, a deterministic reference, and retained negative results are necessary to interpret these gains. The next step is to recover the original simulator and repeat the reward ablation there, followed by independent traffic and geographic validation. The present positive result is a controlled formulation finding, not a deployment or general superiority claim.

APPENDIX A

EXPLICIT COMPARISON WITH THE ORIGINAL TFM

The tables compare the written TFM with the executed implementation. “Not specified” is an evidence boundary, not an assertion about missing source code. All shared changes apply even to our legacy arm. Only reward replacement and training arrival termination are treatment factors. The complete implementation inventory, including all 42 observation indices and event ordering, accompanies the reproducibility package in `docs/25_exact_changes_from_tfm.md`.

TABLE V

ENVIRONMENT, OBJECTIVE, AND EVALUATION COMPARISON. SOURCE LOCATORS REFER TO PRINTED TFM PAGES. ROWS DESCRIBE SHARED CHANGES UNLESS MARKED AS FACTORS.

Item	Original TFM definition	Executed experiment
Radio environment	Barcelona, 2.1 GHz, 100 m, 133 sites, nominal 1 m grid; strongest sector RSS [1, pp. 6–7, Sec. 3.1, Table 2]	Received map retained; 87 station operator retained. No ray tracing regeneration; exact source version and sentinel remain unverified.
Lookup and geometry	3D building propagation geometry; lookup and integer encoding not specified [1, pp. 6–7, Sec. 3.1]	Native coordinate spacing, nearest lookup, float radio arithmetic; –128 treated as absent path. No collision model.
Motion	Scalar speed/selected heading update; five acceleration and eight heading bins [1, p. 9, Eq. (2); pp. 12, 15]	Inertial velocity, continuous radial/lateral acceleration, norm bound 5 m/s^2 , trapezoidal position update, hard speed cap 25 m/s.
Time and speed	1 ms in prose, 0.1 s in table; 2000 timesteps; 100 m/s penalty threshold [1, pp. 14–16, Tables 3–4]	1 s steps, 200 s deadline. New time discretization and speed constraint.
Arrival (factor)	10 m tolerance, stopping intention, no arrival termination [1, p. 9, Sec. 3.2; p. 12, Sec. 5.2; p. 15, Table 3]	Distance $\leq 10 \text{ m}$ and speed $\leq 2 \text{ m/s}$. Training termination varied; every evaluation ends at first safe arrival.
Failure handling	Energy termination; boundary/RSS/speed penalties [1, p. 12, Sec. 5.2, Eq. (14)]	Energy exhaustion, boundary crossing, five consecutive RSS outages terminate. Physical failure precedes arrival; a valid deadline arrival succeeds.
Observations	Position, serving BS, RBG, remaining energy, buffer [1, p. 12, Sec. 5.2]	42 scaled features including goal bearing/distance, velocity, time, consumed energy, prior arrival, phases, serving SINR, candidates, and masks.
Network actions	Station and RBG requests, A3 inequality [1, p. 9, Eq. (3); p. 12, Sec. 5.2]	Stay plus top four RSS stations; $> 3 \text{ dB}$ advantage and $> -96 \text{ dBm}$; five second decisions plus observed outage requests; no time to trigger counters.
Resource allocation	Policy requests RBG; availability fallback; random occupancy of at least half the groups [1, pp. 9, 12, 14–15]	RBG is not learned. First free group, exactly six occupied groups per station in a deterministic pattern; global label phase is not independent load variation.
Rate and interference	Bandwidth plus log of SNR; neighboring RSS sum labeled uplink [1, p. 10, Eqs. (7)–(9)]	Linear downlink cochannel interference, SINR, $B \log_2(1 + \text{SINR})$. Retain 1.44 MHz and -103.41 dBm combined noise; change link interpretation.
Queue and traffic	Poisson formulation, ratio delay, 160 KB, 4800 bit handover overhead; conflicting data packet sizes [1, pp. 9–10, Eqs. (4)–(6); pp. 14–15]	Deterministic 200 kbit/s, decimal 1.28 Mbit cap, same handover overhead. Postservice backlog/rate censored at 200 s; record overflow without terminating.
Energy	Mass/efficiency/lift-to-drag expression; capacity labeled 1000 kW [1, p. 10, Eq. (10); p. 15, Table 3]	Consumed energy integral of $100 + 0.4v^2 \text{ W}$; 100 kJ budget. New uncalibrated proxy, not a conversion of original capacity.
Reward (factor)	Positive reciprocals, binary +12 progress, penalties [1, p. 12, Eqs. (13)–(14); p. 15, Table 3]	Legacy structure uses new common physical inputs and no soft speed penalty. Replacement adds first arrival/terminal failure payments, time cost, bounded radio costs, and terminal corrected potential shaping.
Deterministic reference	Constant speed, closest discrete heading, viability counters [1, pp. 13–14, Sec. 5.4]	Continuous desired velocity tracking with braking; strongest admissible station and shared masks. A new reference, not a reproduction of that greedy policy.
Splits and comparison	PPO/greedy and weight comparisons; separate validation and test sets not specified [1, pp. 15–16, Secs. 6.2–6.3]	4096 training, 64 validation, 200 standard test, 200 longer test routes; five seeds per treatment, frozen budget. Same map, not a geographic holdout.
Reported outcomes	SNR, outage, interference, remaining energy, handovers, example paths [1, pp. 16–20, Sec. 7]	Mission and recorded feasible success first; secondary metrics conditional on success; every seed, paired bootstrap and failure counts retained.

For completeness, our reset sets velocity, queue, consumed energy, counters, and prior arrival to zero, associates with the strongest station, and allocates its first free group. Each step applies the requested admissible handover before motion, evaluates radio after motion, updates queue and energy, checks failure and arrival, and then computes reward. The TFM does not fully specify this event ordering [1, pp. 9–12, Secs. 3.2–5.2]. The direct energy and packet overflow reward terms removed during the original development [1, pp. 20–21, Sec. 7.3] are not treated as part of its final reward. We also do not claim to have reproduced its mass or efficiency model, random traffic, counter based handovers, or learned resource allocation.

TABLE VI

PPO COMPARISON. THE FIRST ELEVEN ROWS COVER EVERY PARAMETER IN THE ORIGINAL TABLE 4 [1, p. 16]. FURTHER IMPLEMENTATION DETAILS ARE NOT FULLY SPECIFIED IN ITS SECS. 5.3 AND 6.2 [1, p. 13; pp. 15–16].

Parameter	Original TFM	Executed experiment
Episodes	300	Fixed interactions, variable number of episodes
Episode timesteps	2000	200, with explicit early terminals
Rollout steps per environment	8	128
Learning rate	$3 \cdot 10^{-5}$	$3 \cdot 10^{-4}$, constant
Batch size	32	512
Discount	0.99	0.99, retained
GAE lambda	0.95	0.95, retained
Policy clip	0.2	0.2, retained
Value coefficient	0.5	0.5 applied to half MSE; total coefficient 0.25 MSE
Entropy coefficient	0.01	0.005
Target KL	0.03	No threshold; log diagnostic KL only
Implementation	Stable Baselines	Native PyTorch hybrid PPO
Parallel lanes and update epochs	Not specified	64 lanes; 8192 samples per rollout; four epochs
Interaction budget	Episodes/timesteps listed	524,288 per policy, 64 rollout updates
Networks	Not specified	Separate actor/critic, each two 64 unit tanh layers
Policy heads	Discrete action vector listed	2D Gaussian plus masked five choice categorical
Initialization and variance	Not specified	Orthogonal weights; log standard deviation initialized -0.5 , clamped to $[-2.3, 0.5]$
Optimizer and gradient cap	Adam in pseudocode	Adam, epsilon 10^{-5} , gradient norm cap 0.5
Likelihood and entropy details	Not specified	Latent Gaussian likelihood after stored action sampling; sum latent Gaussian and categorical entropy
Reward and advantage scaling	Not specified	Running discounted return variance; no reward centering; reward clip $[-10, 10]$; minibatch advantage normalization
Critic and bootstrap details	Not fully specified	Unclipped half MSE; bootstrap only across nonterminal rollout ends, not across mission/deadline terminals
Evaluation and checkpoint rule	Not specified	Gaussian mean and categorical mode; final checkpoint only; common raw evaluation reward for every treatment

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